POLITECNICO
MILANO 1863

Orbital Mechanics Group ID 2541

Academic Year 2025/2026

Person code	Surname	Name	Bachelor
10828687	Contigiani	Valerio	Aerospace Engineering (Polimi, Milano)
10854893	Cristiani	Marcello	Aerospace Engineering (Polimi, Milano)
11179425	Macchia	Bruno	Aerospace Engineering (Unipi, Pisa)
11153986	Mazurek	Krzysztof	Robotics and Automatic Control (WUT, Warsaw)

1. Interplanetary Explorer Mission

1.1 Mission overview

1.1.1 Mission requirements and objectives

The objective of this study is the preliminary analysis of an interplanetary mission departing from Venus, destined to reach asteroid N.147080, exploiting an intermediate Gravity Assist (GA) on Earth to reduce the overall cost in terms of Δv. The imposed temporal requirements define an extremely wide launch window, spanning from January 1, 2030, to January 1, 2060 (30 years).

The primary optimization goal is to minimize the total mission cost Δv_{tot}, defined as the sum of the impulsive velocity changes required for departure, the powered flyby maneuver, and the final orbital insertion: $\Delta v_{tot} = \Delta v_{dep} + \Delta v_{GA} + \Delta v_{arr}$.

1.1.2 Assumptions and Physical Models

The analysis is conducted under the following simplifying assumptions, consistent with a preliminary feasibility study:

- **Patched Conics Method**: Trajectories are modeled as heliocentric Keplerian arcs (patched conics method), assuming that the planetary Spheres of Influence (SOI) are point-like.
- **Impulsive maneuvers**: All velocity changes are considered instantaneous.

1.2 Design Strategy: The Space Pruning Approach

1.2.1 The Curse of Dimensionality

Optimizing an interplanetary trajectory with an intermediate flyby is a problem characterized by three main degrees of freedom (3-DoF): the departure date (t_{dep}), the flyby date (t_{ga}), and the arrival date (t_{arr}). Considering the breadth of the assigned time window (30 years), an exhaustive search approach (Brute Force Grid Search) with a daily temporal resolution would require the evaluation of over 10^9 possible combinations. Such computational cost makes a blind search unfeasible. To address this complexity, a Space Pruning strategy was adopted, based on the physics of the problem and inspired by global optimization methodologies developed by the ESA Advanced Concepts Team [2].

1.2.2 Objective Function Bounding

Before initiating the search algorithm for trajectories with flybys, it was necessary to establish a rigorous criterion to identify and discard non-competitive solutions. According to the Branch and Bound approach described by Addis et al. (2011) [1], search efficiency depends on defining an Upper Bound (UB) for the cost function. The direct Venus-Asteroid mission was chosen as the "Incumbent Solution". As highlighted by Labunsky et al. (1998) [3], the inclusion of an intermediate Gravity Assist is justified only if it lowers the total cost below that of the direct Baseline Mission.

- **Baseline Computation**: A preliminary Lambert search on the direct transfer within the 2030 - (2030 + $3T_{syn}^{VA}$) (See sec. 1.2.3) interval identified a minimum reference cost $\Delta v_{direct} \approx 14.8$ km/s.
- **Pruning Condition**: Consequently, the constraint $UB = 14.8$ km/s was imposed. Following the concept of Admissible Region introduced by Vasile and De Pascale (2006) [4], the algorithm was designed to immediately prune any computation branch where the partial cost (even of just the first leg) exceeds this value.

1.2.3 Search Space Pruning

Having defined the energy limit, the reduction of the temporal domain was achieved by exploiting the periodicity of planetary configurations, as suggested by Izzo (2010) [2].

The Venus $\rightarrow$ Earth leg drives the mission scheduling. Since both orbits are nearly circular and coplanar, optimal geometries repeat cyclically according to the Synodic Period defined as $1/T_{syn} = |1/T_{Venus} - 1/T_{Earth}|$.

Traditionally, an "Anchor Window" strategy involves identifying the first optimal opportunity and propagating it mathematically by integer multiples of T_{syn}. In this study, a generalized variation of this strategy is applied. To account for the orbital eccentricity—which causes slight variations in the exact timing of the windows—and to ensure no feasible opportunities are missed due to theoretical approximations, the algorithm performs a coarse feasibility scan directly over the entire 30-year period, effectively "discovering" the real windows rather than just predicting them.

1.2.4 Feasibility Filter and Window Identification

The results of the coarse feasibility scan are visualized via a scattered porkchop plot (See Figure 1.1), which effectively subdivides the search space into discrete "islands" of opportunity where the cost is lower than the Upper Bound (UB).

A filtering algorithm is then applied to these islands. The algorithm detects the boundaries of the valid data blocks, converting the continuous temporal domain into a set of discontinuous, bounded grids defined by specific minimum and maximum values ($[t_{dep_{min}},\ t_{dep_{max}}]$ and $[ToF_{min},\ ToF_{max}]$) for each identified window.

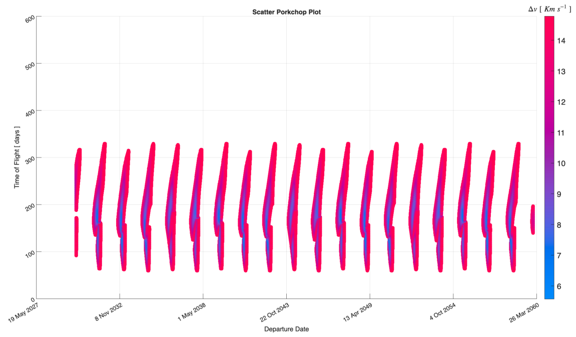


Figure 1.1: Porkchop plot of Venus-Earth leg

1.2.5 Pruned Grid Search

Once the search space has been pruned into specific windows, the grids for the optimization routine are generated.

- **Leg 1 (Venus → Earth)**: Dense grids are generated strictly within the bounds of each identified window.
- **Leg 2 (Earth → Asteroid)**: The grid is defined using the Earth-Asteroid synodic period as a maximum transfer time value.

A nested triple loop then evaluates the total mission cost for every combination within these refined grids. This step identifies the single global minimum candidate among all windows, providing the necessary initial guess for the subsequent local refinement.

1.3 Refined Optimization

1.3.1 Local Grid Search Setup

Following the Pruned Grid Search, the global minimum candidate is identified. However, due to the discretization of the grid, this solution represents a "valley" rather than the precise mathematical minimum. To refine this result and provide an accurate initial guess (x_0) for the gradient-based optimizer, a **Detailed Grid Search** is performed. This involves a high-resolution scan within a narrow temporal window (±20 days) centered around the best candidate found in the pruning phase.

1.3.2 Optimization Solvers

To converge to the precise local minimum, the solution is passed to the `MATLAB` function `fmincon.m`. Unlike grid search methods, which are bound to discrete steps, this solver operates in a continuous domain using a gradient-based approach. The algorithm is initialized with the best result from the Detailed Grid Search and minimizes the objective function while strictly satisfying the imposed non-linear constraints. Specifically, the required precision:

Algorithm	Tol_{Step}	$\text{Tol}_{\text{Constraint}}$	$\text{Tol}_{\text{Function}}$	$\text{Max}_{\text{Func Eval}}$
`SQP`	1×10^{-14}	1×10^{-9}	1×10^{-14}	10000

1.3.3 Mission Objective Function

The objective is to minimize the total mission cost (Δv_{tot}) with respect to the optimization vector $\mathbf{x} = [t_{dep},\ \Delta t_1,\ \Delta t_2]$, representing the departure date and the transfer times for the two legs, where t_{dep} is the departure date (MJD2000), Δt_1 is the Time of Flight for the Venus-Earth leg, and Δt_2 is the Time of Flight for the Earth-Asteroid leg.

1.3.4 Mission Constraint Function

To ensure flight safety and avoid atmospheric drag, a non-linear constraint enforces a minimum pericenter radius $r_p \geq R_E + h_{safe}$, where $R_E = 6371$ km is Earth's radius and $h_{safe} = 300$ km is the altitude margin.

1.4 Mission Analysis Results

1.4.1 Algorithm Setup and Grid Definitions

The optimization algorithm was executed with a total run time of approximately 4 hours. The search space discretization strategies adopted for the different phases of the algorithm are detailed below.

Direct transfer Venus → Asteroid (147080) The baseline for the direct transfer was computed over a departure window spanning from $\mathrm{Dep}_{\mathrm{min}}$ to $\mathrm{Dep}_{\mathrm{min}} + 3\mathrm{T}_{\mathrm{syn}}^{\mathrm{VA}}$. The Time of Flight (ToF) domain was bounded between 100 days and $2\mathrm{T}_{\mathrm{syn}}^{\mathrm{VA}}$.

First Leg Venus → Earth The preliminary feasibility scan for the first leg covered the entire 30-year mission window ($\mathrm{Dep}_{\mathrm{min}}$ to $\mathrm{Arr}_{\mathrm{max}}$). The flight time was constrained between 60 days and one Venus-Earth synodic period ($\mathrm{T}_{\mathrm{syn}}^{\mathrm{VE}}$).

Brute Force Search A coarse global search validated the pruning logic over the following domain: Departure $\in [\mathrm{Dep}_{\mathrm{min}},\ \mathrm{Arr}_{\mathrm{max}} - \mathrm{T}_{\mathrm{syn}}^{\mathrm{VE}}]$, Leg 1 ToF $\in [60,\ \mathrm{T}_{\mathrm{syn}}^{\mathrm{VE}}]$ days, and Leg 2 ToF $\in [60,\ \mathrm{T}_{\mathrm{syn}}^{\mathrm{EA}}]$ days.

Pruned Grid Search (PGS) In this phase, the departure dates and Leg 1 flight times were restricted to the specific "islands" identified by the pruning algorithm (see Sec. 1.2.3). Consequently, the grid bounds vary dynamically for each identified window. The second leg (Earth → Asteroid) was explored with flight times ranging from 60 days to $\mathrm{T}_{\mathrm{syn}}^{\mathrm{EA}}$.

Refined Grid Search The final high-resolution grid search focused on a narrow local neighborhood around the best candidate found in the PGS phase, defined as follows:

Departure	Leg 1 ToF	Leg 2 ToF
$PGS_{Dep} \pm 20$ days	$PGS_{TOF_1} \pm 20$ days	$PGS_{TOF_2} \pm 20$ days

Reference Constants: The search grid relies on the mission window $[\mathrm{Dep}_{\mathrm{min}},\ \mathrm{Arr}_{\mathrm{max}}]$ = [01/01/2030, 01/01/2060] and the synodic periods $T_{syn}^{VE} = 583.92$ d, $T_{syn}^{EA} = 505.06$ d, and $T_{syn}^{VA} = 270.82$ d.

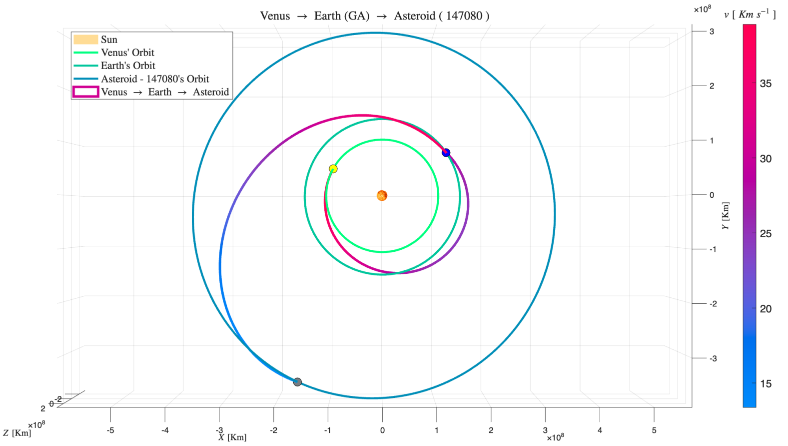
1.4.2 Optimal Trajectory Data

The optimization process successfully converged to a specific mission window. The key epochs and duration of the optimal solution are reported below:

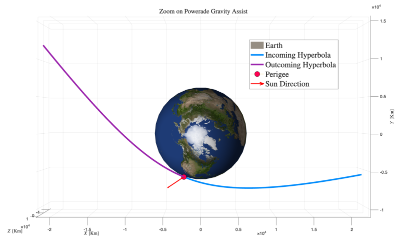
Departure	Earth Flyby	Arrival	Duration	Δt_{fb} (SOI)
27-02-2057 | 23:49:59 (MJD: 20877.49)	26-10-2057 | 15:50:25 (MJD: 21118.16)	13-03-2059 | 06:41:07 (MJD: 21620.78)	**743d 6h 51m 7s**	**73.451h**

1.4.3 Transfer Arcs Characterization

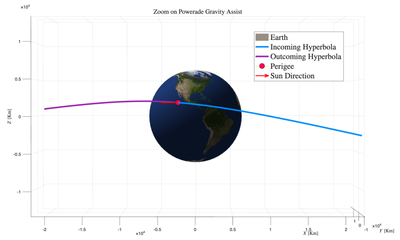
The optimal trajectory resulting from the `fmincon` optimization is illustrated in Figure 1.2. The mission profile consists of two heliocentric arcs connected by a powered gravity assist at Earth.



(a) Full Optimal Mission Trajectory



(b) Zoom at Flyby (View - 1)



(c) Zoom at Flyby (View - 2)

Figure 1.2: Full Heliocentric Trajectory | Zoom at Flyby

1.4.4 Mission Cost Breakdown

The total optimized cost (Δv_{tot}) and its specific components are detailed below:

Maneuver	**Cost [km/s]**
Departure (Venus) (Δv_{dep})	4.04417
Powered Gravity Assist (Earth) (Δv_{GA})	0.52234
Arrival (Asteroid) (Δv_{arr})	4.22099
Total Mission Cost ($\mathbf{\Delta v_{tot}}$)	**8.78750**

1.5 Flyby Analysis

The critical phase of the mission is the Gravity Assist (GA) performed at Earth. This maneuver is modeled using the "Patched Conics" approximation, assuming the spacecraft traverses a hyperbolic trajectory relative to Earth within its Sphere of Influence (SOI).

1.5.1 Geometry of the Hyperbola

The geometry of the flyby is entirely determined by the heliocentric velocity vectors at the discontinuity point between the two legs:

- **Incoming Hyperbolic Excess Velocity ($\vec{v}_\infty^-$):** Derived from the arrival velocity of the first leg relative to Earth ($\vec{v}_{arr,Leg1} - \vec{v}_{Earth}$).
- **Outgoing Hyperbolic Excess Velocity ($\vec{v}_\infty^+$):** Derived from the departure velocity of the second leg relative to Earth ($\vec{v}_{dep,Leg2} - \vec{v}_{Earth}$).

The required turning angle δ, representing the deflection needed for the plane change, is computed via the dot product as $\delta = \arccos\left((\vec{v}_\infty^- \cdot \vec{v}_\infty^+)/(||\vec{v}_\infty^- \cdot \vec{v}_\infty^+||)\right)$.

1.5.2 Powered vs. Unpowered Gravity Assist

In a purely ballistic (unpowered) flyby, the magnitude of the relative velocity is conserved ($||\vec{v}_\infty^-|| = ||\vec{v}_\infty^+||$), and the planet only rotates the velocity vector. However, for this mission, a **Powered Gravity Assist** model was adopted to maximize flexibility. This implies that a propulsive impulse (Δv_{GA}) is applied at the pericenter of the hyperbola to change the energy of the orbit. The cost of this maneuver is calculated as the difference between the required velocity at pericenter to exit on the second leg (v_p^+) and the natural velocity reached at pericenter from the first leg (v_p^-), expressed as $\Delta v_{GA} = |v_p^+ - v_p^-| = \left|\sqrt{||\vec{v}_\infty^+||^2 + 2\mu_E/r_p} - \sqrt{||\vec{v}_\infty^-||^2 + 2\mu_E/r_p}\right|$.

The solution gives a ratio of 0.0676 (6.76%) between the pericenter impulse and the total flyby Δv. This result confirms the maneuver's high efficiency.

Moreover, since the maneuver is executed in proximity of the Line of Nodes of the asteroid's orbit, the cost of this phase is minimized.

1.5.3 Operational Constraints Verification

Post-optimization analysis confirms that the trajectory respects the safety limit against atmospheric drag, with a pericenter altitude $h_p = r_p - R_E \geq 300$ km, ensuring physical feasibility and avoiding thermal damage.

1.6 Conclusions

This study successfully demonstrated the feasibility and efficiency of a Venus-to-Asteroid mission exploiting an intermediate Earth Gravity Assist. The primary objective of minimizing the total mission cost (Δv_{tot}) was achieved through a rigorous "Space Pruning" strategy, which effectively addressed the "Curse of Dimensionality" inherent in a 30-year search window.

The analysis yielded the following key conclusions:

- **Effectiveness of Space Pruning:** The hierarchical reduction of the search space—from the global direct-transfer baseline to synodic window identification and finally to pruned grid searches—allowed for the efficient evaluation of millions of potential trajectories. This approach successfully isolated the global optimum without necessitating a prohibitive brute-force computation.

- **Benefit of the Gravity Assist:** The comparison between the direct transfer baseline ($\approx$ 14.8 km/s) and the optimized flyby trajectory ($\Delta v_{tot} \approx 8.8$ km/s) confirms the significant energetic advantage of the Earth flyby. The inclusion of the intermediate leg reduced the total impulse requirement by approximately **6 Km/s**.

- **Optimization Refinement:** The transition from discrete grid searches to the continuous gradient-based solver (`fmincon`) proved crucial. The local optimizer successfully refined the solution, pinpointing the precise epoch times and reducing the residual cost to the mathematical local minimum.

- **Operational Feasibility:** The final trajectory satisfies all imposed physical constraints. Specifically, the powered gravity assist at Earth respects the minimum altitude safety margin ($h_p \geq 300$ km), ensuring that the maneuver is not only theoretically efficient but also operationally viable in terms of atmospheric drag avoidance. Specifically, the final and optimal altitude h_p is slightly above 300 Km.

In conclusion, the identified trajectory represents a robust preliminary mission profile, offering a high-efficiency alternative to direct transfers and validating the use of pruning algorithms for complex interplanetary mission design.

2. Planetary Explorer Mission

2.1 Introduction

2.1.1 Mission Overview

The PoliMi Space Agency has proposed a Planetary Explorer Mission aimed at performing detailed observations of celestial bodies within the Solar System. As part of the mission analysis team, the objective of this study is to carry out a comprehensive orbit analysis and ground track estimation for the spacecraft. The project requires characterizing the nominal orbit, evaluating the effects of gravitational perturbations, comparing different numerical propagation methods, and designing an orbit modification to achieve a repeating ground track for targeted observations.

2.1.2 Selected Scenario

Among the proposed mission profiles, this report focuses on the Departure Planet Observation scenario. Before the injection into the interplanetary transfer trajectory, the spacecraft is tasked with an observation campaign of the departure planet. For this specific mission analysis, the central body is Venus. This choice implies that the gravitational environment is dominated by Venus' gravitational parameter (μ_V) and its specific zonal harmonic coefficients (mainly J_2), while the primary third-body perturbation source is the Sun.

2.2 Nominal Orbit Analysis

2.2.1 Initial Keplerian Elements

The analysis begins with the propagation of the nominal orbit starting from a defined set of Keplerian elements. The motion is described in a Venus-Centred Inertial (VCI) Equatorial frame, where the fundamental plane coincides with Venus' equator and the principal direction points towards the vernal equinox equivalent for Venus at the reference epoch.

The initial epoch for the simulation is set to March 18, 2053, at 07:27:51. The initial orbital parameters assigned for the mission are:

a [km]	**e** [-]	**i** [deg]	$\boldsymbol{\Omega}$ [deg]	$\boldsymbol{\omega}$ [deg]	$\boldsymbol{\theta_0}$ [deg]
181 712	0.2083	133	143	60	0

It is important to note that the initial true anomaly is set to $\theta_0 = 0°$, placing the spacecraft at the pericenter of its orbit at the start of the simulation.

2.2.2 Nominal Orbit Visualization

The nominal orbit, propagated assuming a Keplerian two-body problem (unperturbed motion), is visualized in Figure 2.1. Given the semi-major axis of approximately 1.8×10^5 km, the spacecraft operates in a high-altitude orbit. The orbital period T, calculated via the relation $T = 2\pi\sqrt{a^3/\mu_V}$, corresponds to approximately 9.883 days. This long period suggests that the spacecraft will spend a significant portion of its time at high altitudes, potentially exposing it to third-body perturbations from the Sun more intensely than a Low Venus Orbit (LVO).

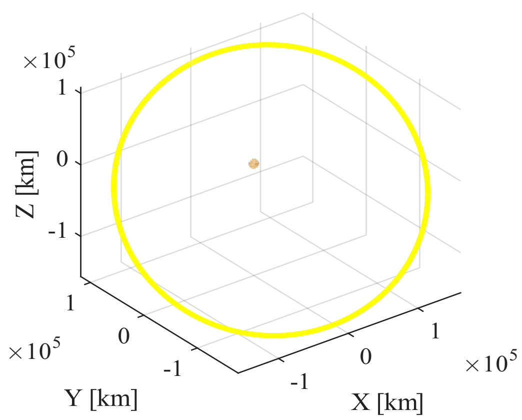


Figure 2.1: Nominal Venus-centred orbit representation in the VCI frame.

2.3 Perturbation Modeling

To accurately predict the long-term evolution of the spacecraft's orbit, the relevant perturbing accelerations have been modeled and included in the equations of motion. Specifically, the non-spherical mass distribution of Venus and the gravitational influence of the Sun have been considered.

2.3.1 Venus Zonal Harmonics (J_2)

The oblateness of Venus is modeled by considering the second zonal harmonic coefficient, J_2. The corresponding perturbing acceleration $\boldsymbol{a}_{J2}$ is expressed in the Cartesian frame. Since zonal harmonics depend only on latitude (and distance) and are independent of longitude, the formula can be applied directly using the position vector $\boldsymbol{r} = x\hat{\boldsymbol{i}} + y\hat{\boldsymbol{j}} + z\hat{\boldsymbol{k}}$ in the inertial frame, provided the z-axis is aligned with the planet's rotation axis.The acceleration components are given by:

$$\boldsymbol{a}_{J2} = \frac{3}{2}\frac{J_2 \mu_V R_V^2}{r^5}\left[\left(5\frac{z^2}{r^2} - 1\right)x\hat{\boldsymbol{i}} + \left(5\frac{z^2}{r^2} - 1\right)y\hat{\boldsymbol{j}} + \left(5\frac{z^2}{r^2} - 3\right)z\hat{\boldsymbol{k}}\right]$$

where: μ_V is Venus' gravitational parameter, R_V is Venus' mean equatorial radius, $r = \sqrt{x^2 + y^2 + z^2}$ is the distance from the planet center,

2.3.2 Sun Third-Body Perturbation

The gravitational attraction exerted by the Sun is modeled as a third-body perturbation. The acceleration acting on the spacecraft relative to Venus, $\boldsymbol{a}_{3B}$, is calculated as the difference between the acceleration induced by the Sun on the spacecraft and the acceleration induced by the Sun on Venus:

$$\boldsymbol{a}_{3B} = \mu_{Sun}\left(\frac{\boldsymbol{r}_{Sun} - \boldsymbol{r}}{|\boldsymbol{r}_{Sun} - \boldsymbol{r}|^3} - \frac{\boldsymbol{r}_{Sun}}{|\boldsymbol{r}_{Sun}|^3}\right)$$

where: μ_{Sun} is the Sun's gravitational parameter, $\boldsymbol{r}_{Sun}$ is the position vector of the Sun with respect to Venus, $\boldsymbol{r}$ is the position vector of the spacecraft with respect to Venus.

2.4 Ground Track Analysis

This section characterizes the ground track coverage and evaluates the feasibility of a repeating orbit. The analysis is performed first under ideal Keplerian conditions and subsequently including J_2 and Solar perturbations. The observation windows were selected as:

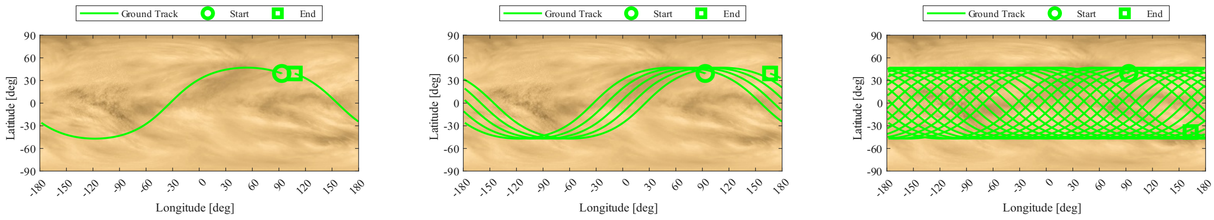
1. Single Orbit ($1T$): To visualize the fundamental closed shape of the track.
2. Multiple Orbits ($5T$): To highlight the longitudinal spacing between consecutive passes.
3. Full Venus Rotation (243 days): To assess the global coverage achievable in one Venusian sidereal day.

For the repeating orbit, a ratio $k : m = 34 : 1$ was selected, yielding a semi-major axis $a_{rep} \approx$ 146409 km ($T_{rep} \approx 7.15$ days).

2.4.1 Unperturbed Scenario

- Nominal Orbit ($a = 181712$ km): Figure 2.2 shows the natural Eastward drift of $\approx 14.6°$ per orbit caused by Venus' retrograde rotation. The track covers the planet densely over 243 days but does not repeat.

- Repeating Orbit ($a_{rep} = 146409$ km): The designed orbit behaves perfectly. As illustrated in Figure 2.3, the track closes exactly upon itself after one Venusian sidereal day, confirming the validity of the $k = 34$ design.



(a) Single orbital period (b) Five orbital periods (c) Full Venusian sidereal day

Figure 2.2: Nominal Ground Track (Unperturbed Scenario)

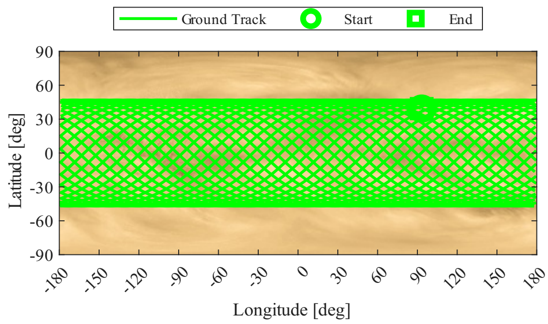
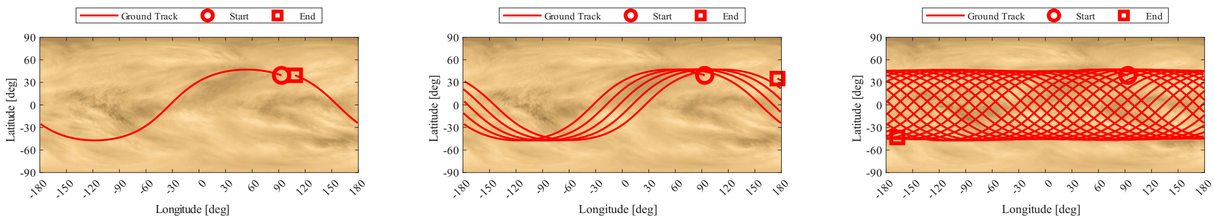


Figure 2.3: Repeating Ground Track (Unperturbed Scenario)

2.4.2 Perturbed Scenario

When the assigned perturbations (J_2 + Sun) are included, the orbital elements evolve over time, breaking the symmetry of the ground track.

- Nominal Orbit: The perturbations induce secular and long-period variations that modify the nodal period. While the global coverage pattern remains qualitatively similar to the unperturbed case, the exact crossing points shift over time (see Figure 2.4).
- Repeating Orbit Analysis: The strict repetition condition does not hold under perturbations. As shown in Figure 2.5, the track fails to close after 243 days. This occurs because the repeating condition relies on a constant nodal period. However, J_2 induces a secular precession of the RAAN ($\dot{\Omega}$) and perigee ($\dot{\omega}$), while the Solar third-body force adds periodic variations. These effects alter the "effective" period of the satellite relative to the rotating planet, causing a cumulative longitudinal drift ($\Delta\lambda_{drift}$) that prevents the track from overlapping with the initial pass.



(a) Single orbital period (b) Five orbital periods (c) Full Venusian sidereal day

Figure 2.4: Nominal Ground Track (Perturbed Scenario)

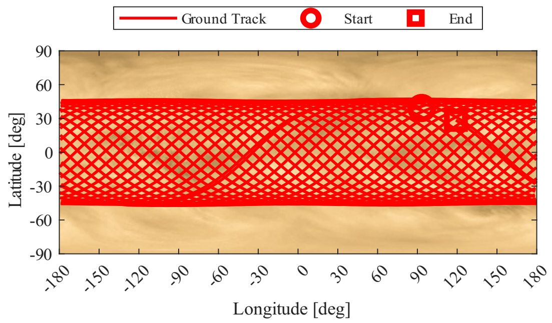


Figure 2.5: Repeating Ground Track (Perturbed Scenario)

2.5 Elements Propagation and Behaviour

Up to this section, the trajectory analysis (e.g., Ground Tracks) was performed using Cartesian propagation, which is ideal for visualizing the spacecraft's position in space. However, to analyze the long-term evolution of the orbit and isolate the effect of perturbations, it is necessary to introduce the Gauss Planetary Equations.

2.5.1 Numerical Methods Comparison

- **Cartesian Propagation (Cowell's Method)**: This method involves the direct numerical integration of the equations of motion in the inertial frame ($\ddot{\boldsymbol{r}} = -\frac{\mu}{r^3}\boldsymbol{r} + \boldsymbol{a}_{pert}$). While conceptually straightforward, it requires converting the state vector back to Keplerian elements at each time step to analyze the orbital geometry.

- **Gauss Planetary Equations**: This approach integrates the rates of change of the orbital elements ($\dot{a}$, $\dot{e}$, $\dot{i}$, Ω, ω, θ) directly. This method is introduced, since Gauss equations provide immediate physical insight into how specific forces affect the orbit shape and orientation. Furthermore, calculating the variations of elements directly is more suitable for the subsequent long-term analysis, where mean elements will be extracted to study secular trends.

2.5.2 Validation

Before proceeding with the long-term analysis using Gauss equations, a validation step is performed to ensure consistency with the established Cartesian baseline. The absolute difference $|\alpha_{Gauss} - \alpha_{Cart}|$ was computed for all six orbital elements over one Venusian year.
Figure 2.6 illustrates the results. The error remains extremely low and bounded for all parameters:

- **Semi-major axis (a)**: The discrepancy remains bounded below 10^{-6} km (millimeter scale). Since a is directly linked to the orbital energy, this negligible error confirms that both propagators maintain energy consistency.

- **Eccentricity and Angles**: Eccentricity and angles exhibit errors in the order of 10^{-10} to 10^{-14}, attributable to machine precision. This confirms that the Gauss implementation is numerically equivalent to the Cartesian one, validating its use for the Long-Term Evolution analysis in the next section.

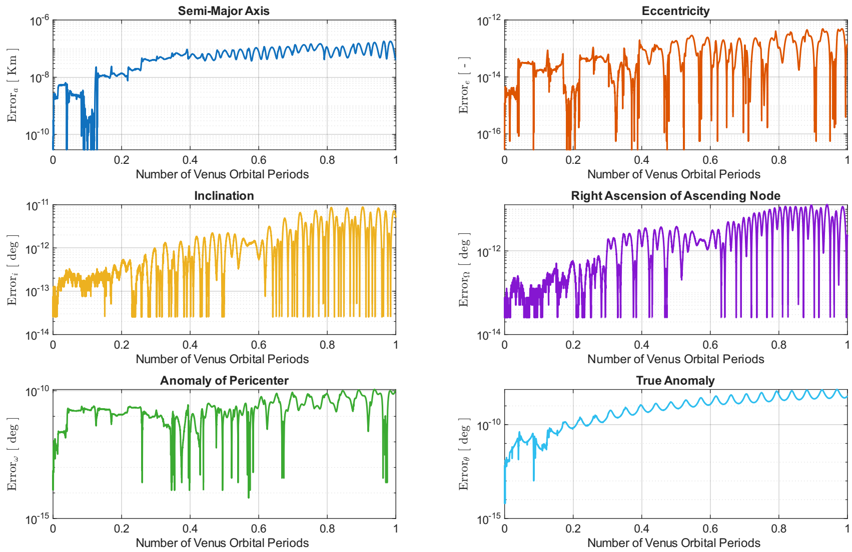


Figure 2.6: Errors | Cartesian vs. Gaussian

2.5.3 Orbital Elements Evolution Analysis

The history of the Keplerian elements was analyzed to isolate the effects of the different perturbations. The simulation covers a period of one Venusian year ($T_{Venus} \approx 225$ days), sufficient to observe both secular trends and long-period solar effects.

Figure 2.7 displays the evolution of the orbital elements, comparing the effect of J_2 only, Sun Third-Body only and the Total perturbation.

- **Semi-major axis** (a): The total variation remains stable around the nominal value, confirming that no orbital decay occurs (conservative forces only). J_2 induces only short-period oscillations, while the Sun adds a minor periodic variation.
- **Eccentricity** (e) **and Inclination** (i): No secular trend is observed. The evolution is dominated by the Solar perturbation, which drives long-period oscillations (period $\approx 1/2 T_{Venus}$), visible in the overlap between the "3B" and "Total" curves.
- **RAAN** (Ω) **and Perigee** (ω): The evolution is dominated by the J_2 secular drift. Ω exhibits a constant nodal regression, while ω precesses linearly. The Solar perturbation adds only a small periodic modulation on top of this trend.
- **True Anomaly** (θ): Being the fast variable, it cycles $0 \rightarrow 360°$ rapidly. The perturbations induce minor phase shifts relative to the unperturbed prediction.

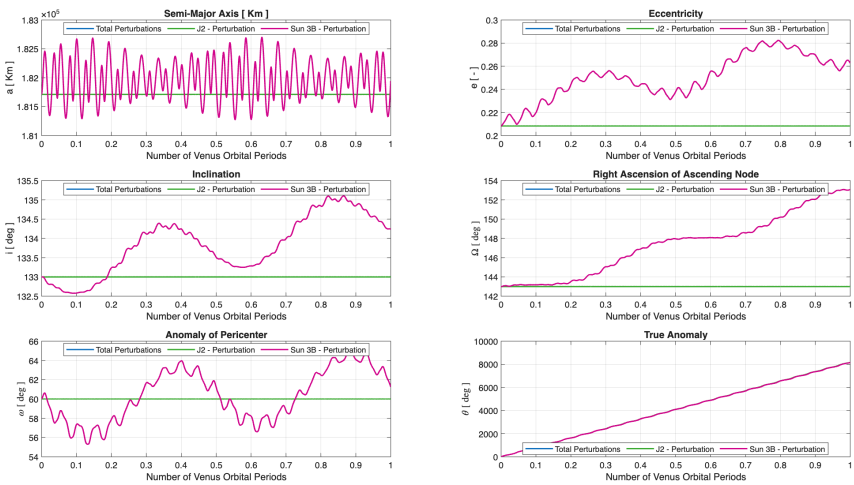


Figure 2.7: All Perturbations | J2 vs. Sun-3B vs. Total

2.6 Orbit Evolution

Gravitational perturbations induce secular variations in the orbital elements. To appreciate the magnitude of these effects and the resulting shift in the trajectory (e.g., nodal precession), the orbit is propagated and visualized over 100 periods:

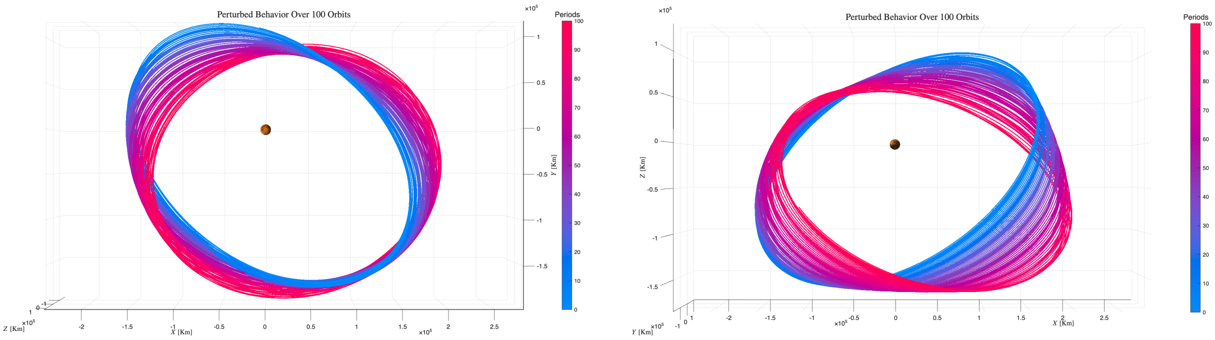


Figure 2.8: Perturbed Orbit | 100 Periods

2.7 High Frequencies Filtering

Orbital perturbations affect the Keplerian elements on different time scales. The time history of any element can be viewed as the superposition of three distinct components:

- **Short-period variations**: High-frequency oscillations occurring within a single orbital period (T), driven primarily by the zonal harmonics of the central body (e.g., J_2).
- **Long-period variations**: Oscillations with periods significantly longer than T, typically related to the rotation of the line of apsides or the motion of disturbing third bodies.
- **Secular Variations**: Monotonic trends that persist over very long time spans.

To isolate the long-term and secular behavior required for the mission analysis, it is necessary to filter out the high-frequency components. This is achieved by applying a low-pass filter to the time series of the orbital elements. In this analysis, a moving average filter (implemented via the `MATLAB` function `movmean`) is employed. The critical design parameter for this filter is the window size (T_w), which determines the cut-off frequency.

2.7.1 Selection Of Filtering Windows

The choice of the window size is physically motivated by the periodicity of the perturbations to be suppressed.

- **Suppression of J_2 Effects (Short-Period)**: The J_2 perturbation depends on the satellite's position relative to the equator and repeats cyclically with the satellite's true anomaly. Consequently, the fundamental frequency of these variations corresponds exactly to the satellite's orbital period (T_{sat}). To remove these short-period oscillations and reveal the underlying evolution, the filtering window is set to match the nominal orbital period of the spacecraft: $T_{w,J2} = T_{sat}$. This specific window width is chosen because short-period effects average out over exactly one complete revolution (as the true anomaly sweeps from 0 to 2π). Therefore, a window of a single period is the most efficient choice to smooth the signal without masking the physical long-term dynamics.
- **Suppression of Third-Body Effects (Long-Period)**: Perturbations due to a third body (the Sun) vary with the relative geometry between the planet (Venus) and the disturbing body. This geometry repeats with the period of Venus's revolution around the Sun ($T_{VenusYear}$).To filter out these long-period oscillations and isolate the pure "Secular"

trend, a second, wider window is selected corresponding to the period of the perturbing body: $T_{w,3rdBody} \approx T_{VenusYear}$. Applying this filter removes the annual oscillations caused by the Sun, leaving only the secular drift of the orbital elements.

2.7.2 Comparison Between Filtered and Un-Filtered Elements

Figure 2.9 illustrates the effectiveness of the filtering process.

- **Semi-Major Axis (a)**: The unfiltered signal (Blue) shows high-frequency noise. However, the filtered evolution (Pink) is almost perfectly constant. This aligns with theory, as J_2 does not cause secular variations in the semi-major axis, confirming the filter has correctly removed the periodic noise.
- **Eccentricity (e) and Inclination (i)**: The filtering removes the "thickness" of the line (short-period oscillations), clearly revealing the large-scale sinusoidal exchange of angular momentum driven by the third-body perturbation (Sun).
- **RAAN (Ω) and Pericenter (ω)**: The filtered lines show clear secular drifts (linear trends) combined with long-period modulations, which would be difficult to quantify in the noisy unfiltered data.
- **True Anomaly (θ)**: As the "fast variable" of the system, its evolution is dominated by the mean motion. The linear trend remains unchanged after filtering, simply reflecting the continuous revolution of the satellite around the central body.

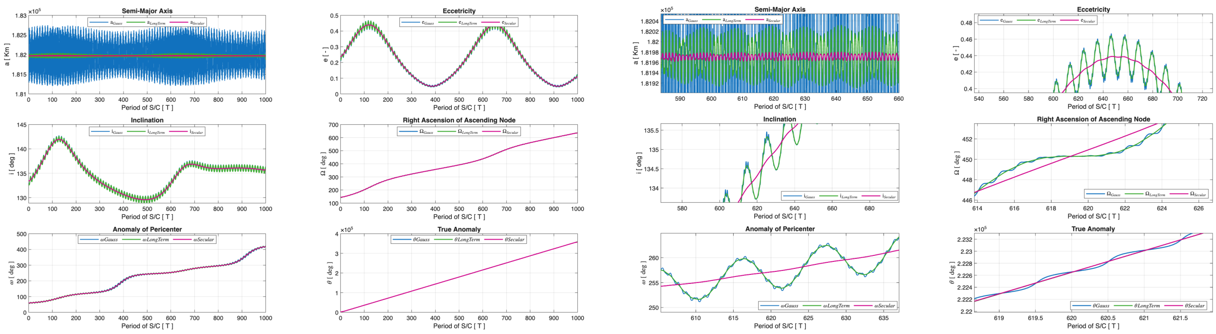


Figure 2.9: Filtering | Filtered vs. Unfiltered

2.8 Real Data Analysis

To validate the perturbation analysis in a real-world scenario, it is convenient to shift focus to objects orbiting in Earth's gravitational field.

2.8.1 LEO Object Analysis | SL-3 R/B

To analyze the perturbation environment in the Low Earth Orbit (LEO) regime, a spent rocket body has been selected. Rocket bodies are ideal candidates for this analysis due to their typically high Area-to-Mass ratio (A/m), which amplifies the effects of non-conservative forces.

Name	NORAD ID	Type	Date Range	Step
`SL-3 R/B`	13819	Rocket Body	1983-02-17 → 1989-10-09	3h

The object follows a nearly circular orbit with a mean perigee radius $r_p \simeq 6961$ km ($h_p \approx 590$ km) and a mean apogee radius $r_a \simeq 7053$ km ($h_a \approx 682$ km). The eccentricity is low ($e \simeq 0.0066$).

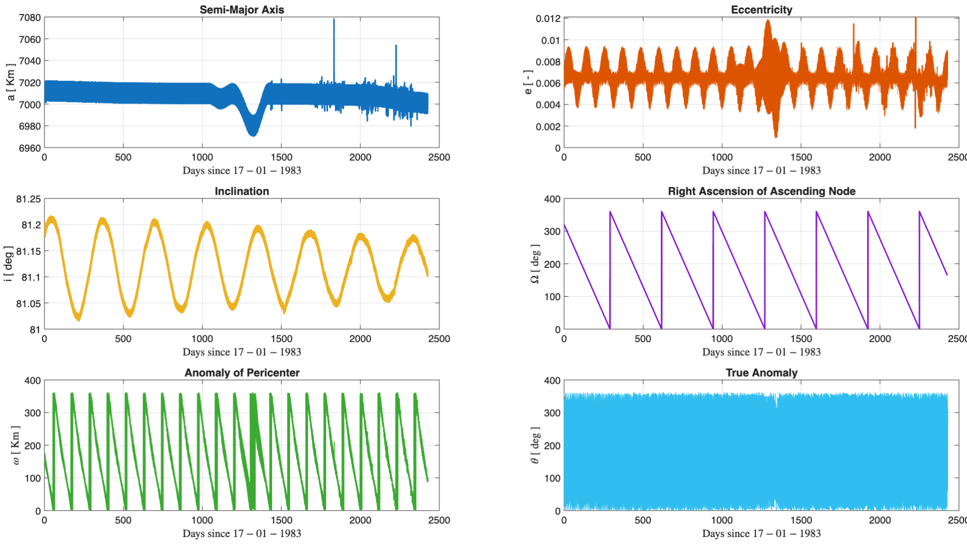


Figure 2.10: Evolution of Keplerian Elements for SL-3 R/B (LEO)

Analysis of Observed Perturbations: The evolution confirms the theoretical predictions for LEO:

1. **Atmospheric Drag:** Secular decay of semi-major axis a. Variations in slope correlate with the 6-year solar cycle activity.
2. J_2 **Effect:** Secular trends in angular elements:
   - **Nodal Regression ($\dot{\Omega} < 0$):** Westward shift of the Line of Nodes ($i < 90°$).
   - **Apsidal Precession ($\dot{\omega} \neq 0$):** Rotation of the argument of perigee over time.

2.8.2 MEO Object Analysis | Titan 3C Transtage R/B

For the second case study, an object in a high-altitude orbit (Sub-Synchronous / Super-MEO) has been chosen to observe the dynamics where drag is negligible.

Name	NORAD ID	Type	Date Range	Step
`Titan 3C Transtage R/B`	11147	Rocket Body	1984-01-01 → 1991-01-01	3h

The orbit is characterized by mean radii $r_p \simeq 36513$ km and $r_a \simeq 38843$ km. Although the eccentricity is low ($e \simeq 0.026$), the altitude ($\sim 30,000$ km) places this object well above the GPS constellation.

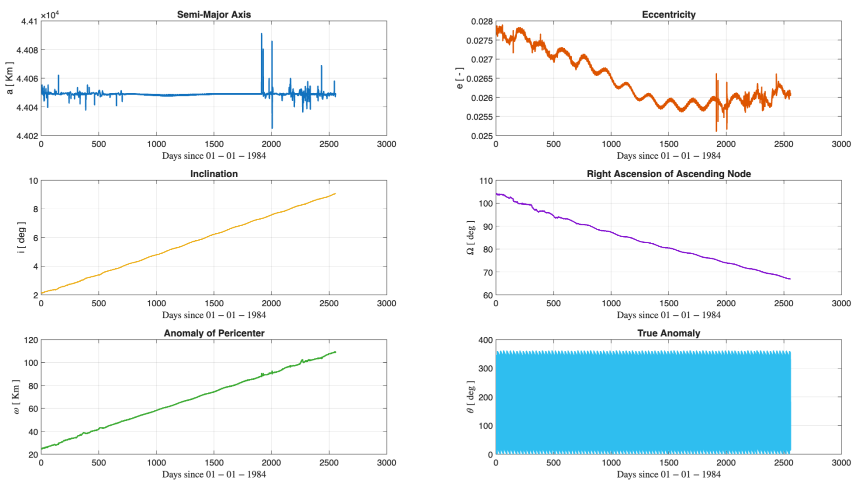


Figure 2.11: Evolution of Keplerian Elements for Titan 3C Transtage R/B (MEO)

Analysis of Observed Perturbations: Hierarchy of perturbations differs from LEO:

1. **Absence of Drag:** a remains constant (density is effectively zero).
2. **Third-Body (Dominant):** Sun and Moon drive dynamics as J_2 decays ($r^{-3.5}$).
3. **Lunisolar Cycles:** Long-period oscillations observed in e and i, unlike the stable inclination in LEO.

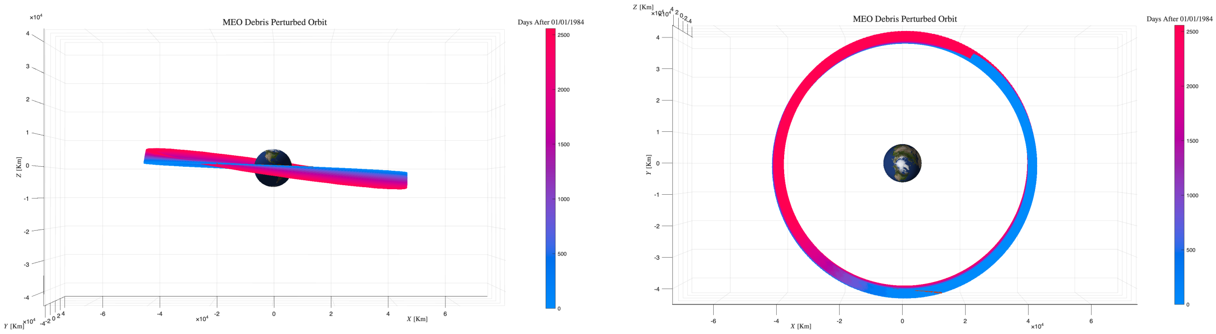


Figure 2.12: MEO Orbit

2.8.3 Conclusion: Third-Body Analysis

The gravitational influence of the Moon and Sun varies by regime. **In High-Altitude (Titan 3C Transtage R/B)**, third-body effects are significant, causing visible long-period oscillations in e and i and contributing to ω drift. **In LEO (SL-3 R/B)**, Earth's gravity dominates overwhelmingly. No long-term third-body effects are visible; dynamics are driven purely by J_2 and Atmospheric Drag.

References

[1] B. Addis, A. Cassioli, and M. Locatelli. A global optimization method for the design of space trajectories. *Computational Optimization and Applications*, 48(3):635–652, 2011. doi: 10.1007/s10589-009-9261-6.

[2] D. Izzo. Global optimization and space pruning for spacecraft trajectory design. In B. A. Conway, editor, *Spacecraft Trajectory Optimization*, chapter 7, pages 178–201. Cambridge University Press, 2010.

[3] A. V. Labunsky, O. V. Papkov, and K. G. Sukhanov. *Multiple Gravity Assist Interplanetary Trajectories*. CRC Press, 1998.

[4] M. Vasile and P. De Pascale. Preliminary design of multiple gravity-assist trajectories. *Journal of Spacecraft and Rockets*, 43(4):794–805, 2006.